

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Model Question Paper Solutions (All 5 Modules)

Module 1

Q.01 a. Explain the Platform Evolution of different computer technologies with a neat diagram.

Platform Evolution

Computer systems have developed over five generations, each covering around 10 to 20 years. The change from one generation to another was not immediate. Usually, there was an overlap of about 10 years between two generations as new technologies slowly replaced the old ones.

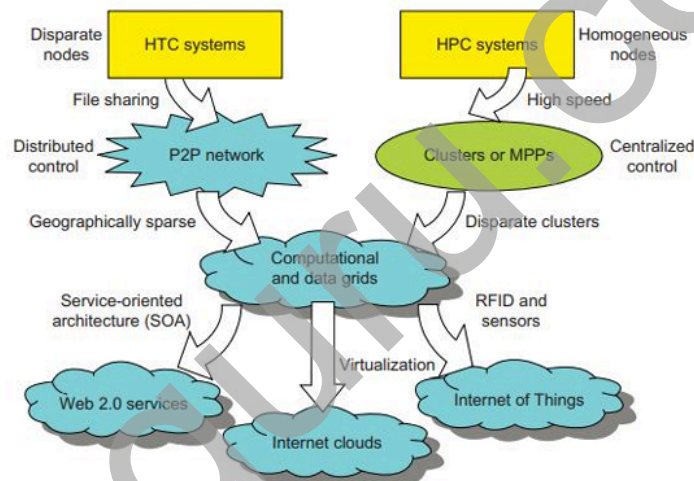


FIGURE 1.1

Evolutionary trend toward parallel, distributed, and cloud computing with clusters, MPPs, P2P networks, grids, clouds, web services, and the Internet of Things.

Generation-wise Platform Evolution

1. Mainframe Era (1950–1970):

- Designed for use by large organizations and government agencies.
- Examples include IBM 360 and CDC 6400 systems.
- Featured centralized, powerful computers with limited user access.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

2. Minicomputer Era (1960–1980):

- Offered lower-cost computing solutions for smaller companies and educational institutions.
- Examples: DEC PDP 11 and VAX series.
- More affordable and interactive compared to mainframes.

3. Personal Computer (PC) Era (1970–1990):

- Marked by the introduction of VLSI-based microprocessors.
- Made computing accessible in homes, schools, and offices.
- Enabled widespread personal computing.

4. Portable Devices Era (1980–2000):

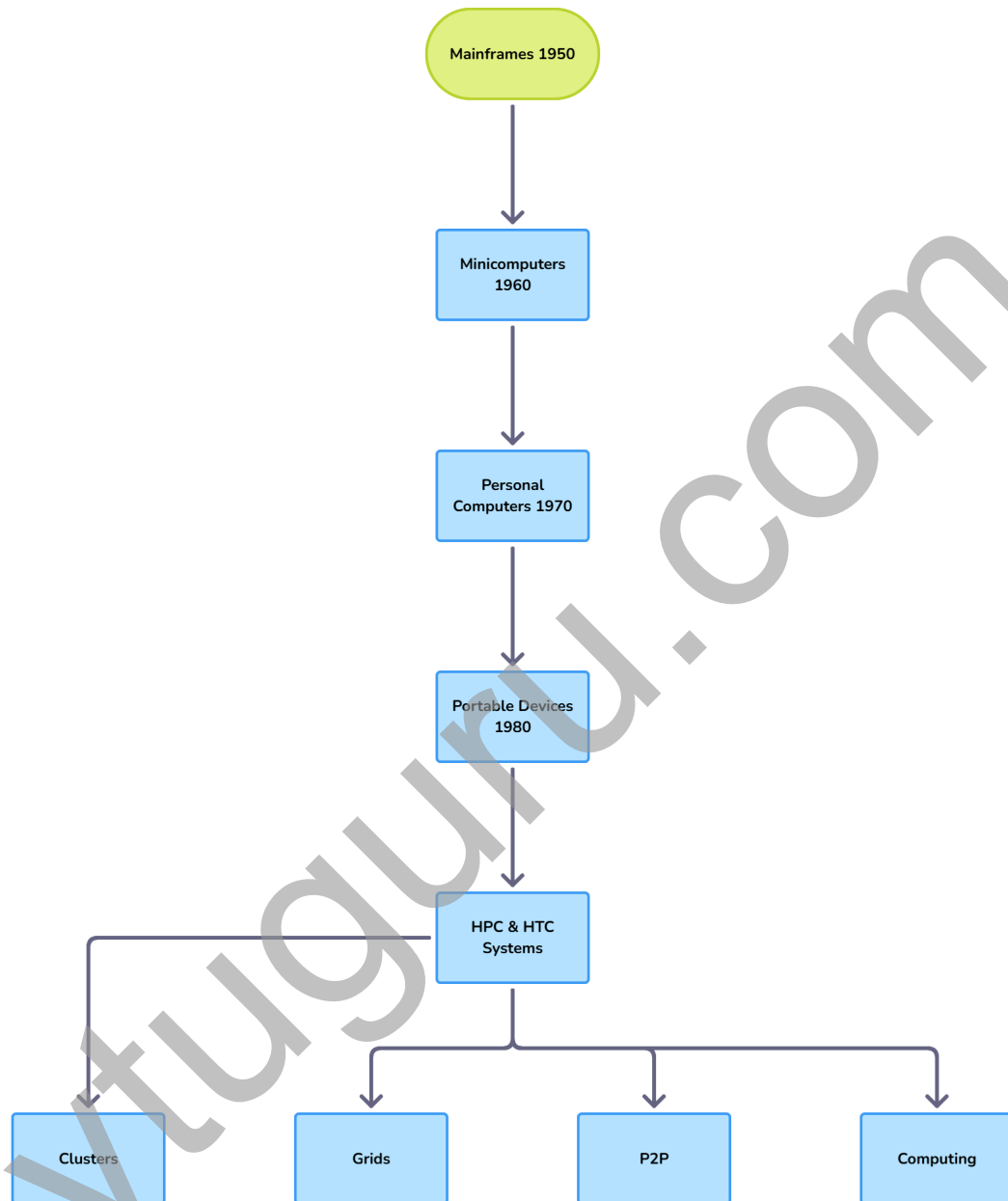
- Growth in the use of laptops, PDAs, and other wireless devices.
- Allowed mobile and anytime-anywhere computing.
- Supported the idea of ubiquitous access to computing resources.

5. HPC and HTC Era (1990–Present):

- Focus on High-Performance Computing (HPC) and High-Throughput Computing (HTC).
- Includes technologies like clusters, grids, and cloud computing.
- Supports large-scale scientific research and commercial web applications.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University



Current Trends in Computing

- Emphasis on accessing shared resources through the web and handling large-scale data sets.
- Traditional supercomputers using Massively Parallel Processing (MPP) are being replaced by clusters of similar nodes.

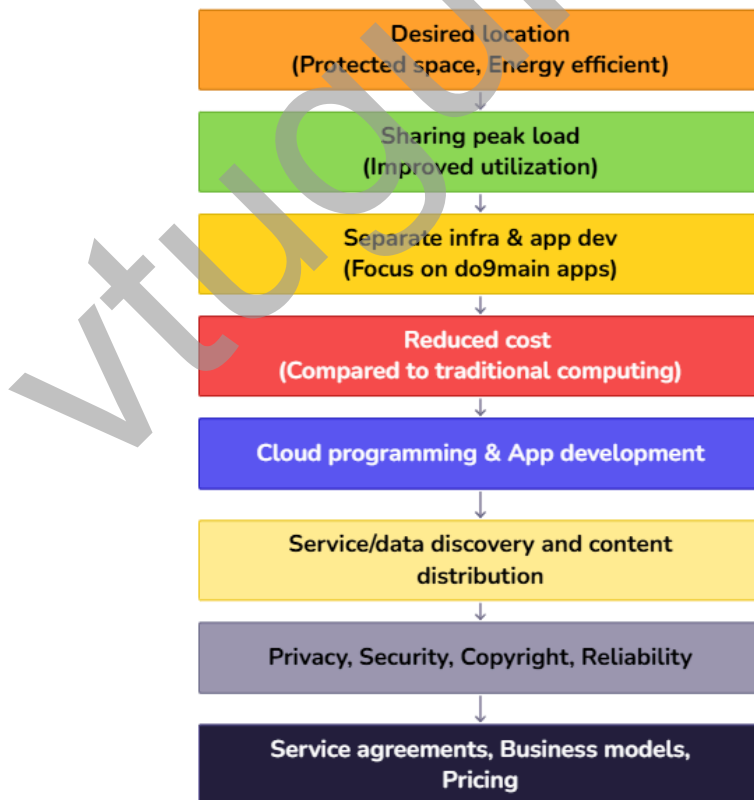
BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- High-Throughput Computing (HTC) systems focus on peer-to-peer (P2P) resource sharing and delivering services via cloud and web platforms.

Q.01 b. Outline eight reasons to adapt the cloud for upgraded Internet applications and web services.

- Prefer locations with secure environments and better energy efficiency.
- Enable sharing of peak loads across many users to boost overall resource use.
- Separate tasks of maintaining infrastructure from developing specific applications.
- Offer much lower costs compared to traditional computing methods.
- Support programming and development of cloud-based applications.
- Facilitate discovery of services and data, along with distributing content effectively.
- Address concerns about privacy, security, copyrights, and system reliability.
- Define service contracts, pricing models, and business strategies.



BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Q.01 c. Briefly explain Message Passing Interface (MPI).

- A standardized library that enables communication between different processes in parallel computing environments.
- Widely used on supercomputers and clusters to perform high-performance computing tasks.
- Each process runs independently but shares data by exchanging messages.
- Provides functions such as MPI_Send and MPI_Recv to handle sending and receiving of messages between processes.

Q.02 a. Summarize VM Primitive Operations with relevant diagram.

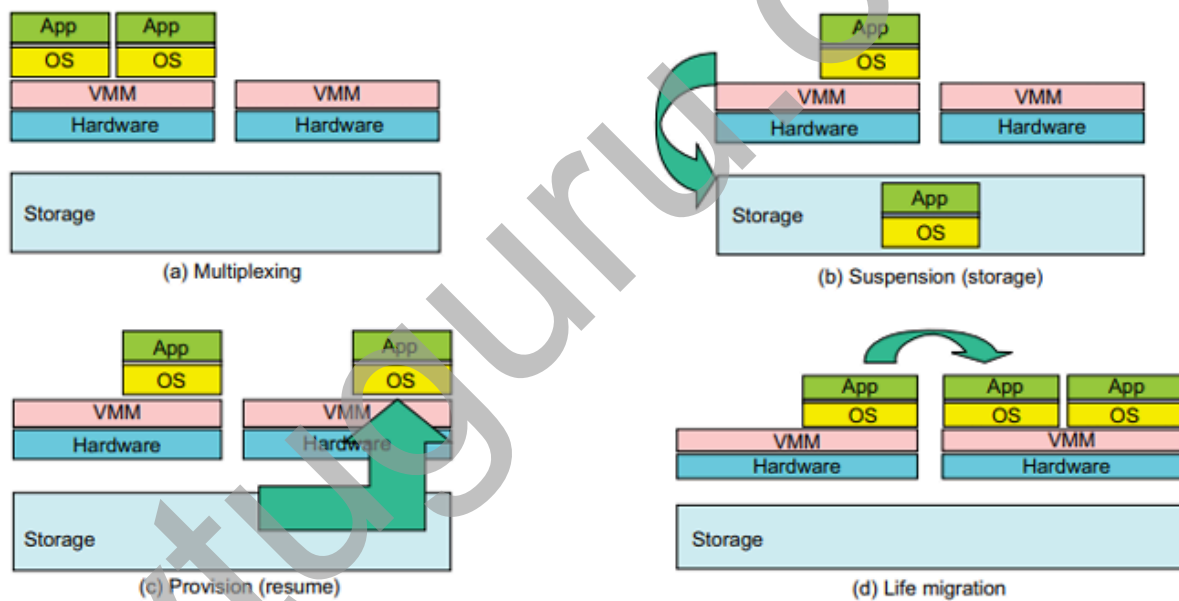


FIGURE 1.13

VM multiplexing, suspension, provision, and migration in a distributed computing environment.

(Courtesy of M. Rosenblum, Keynote address, ACM ASPLOS 2006 [41])

Virtual Machine Monitor (VMM) and Virtual Machines

- The Virtual Machine Monitor (VMM) provides a virtual environment to the guest operating system that mimics real hardware.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- In full virtualization, the VMM offers a virtual machine that appears identical to a physical machine.
- This setup allows standard operating systems like Windows 2000 or Linux to run just as they would on actual hardware.
- Mendel Rosenblum described the core low-level operations of a VMM.

Basic VM Operations:

1. A virtual machine can be shared across multiple physical hosts (multiplexing).
2. A VM can be paused or suspended and its state saved to persistent storage.
3. A suspended VM can be resumed later or moved to a different hardware platform.
4. A VM can be migrated from one physical machine to another while running.

Advantages of Using VMs:

- Virtual machines can run on any compatible hardware.
- Makes moving distributed applications between systems easier.
- Improves overall server resource utilization.
- Multiple server workloads can be consolidated onto a single physical machine.
- Reduces the number of physical servers needed (avoiding server sprawl).
- VMware has shown this approach can raise server utilization rates from 5–15% up to 60–80%.

Q.02 b. Illustrate Various system attacks and network threats to the cyberspace, resulting in 4 types of losses with a neat diagram.

Threats to Systems and Networks

- Clusters, grids, clouds, and peer-to-peer (P2P) systems need strong security measures to maintain user trust.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- Network-based viruses can seriously damage routers and servers.
- Such attacks often lead to significant financial losses for businesses and governments.
- Leaked information can compromise data confidentiality.
- Data integrity can be threatened by unauthorized changes, Trojan horse malware, and spoofing.
- Denial of Service (DoS) attacks can disrupt normal system operations and cut off Internet access.
- Weak or missing authentication allows attackers to misuse systems.
- Open environments like data centers and P2P networks are especially vulnerable to attacks.
- Attacks can harm computers, networking equipment, and storage devices.
- Problems in routers and gateways can reduce the overall trust in public network services.

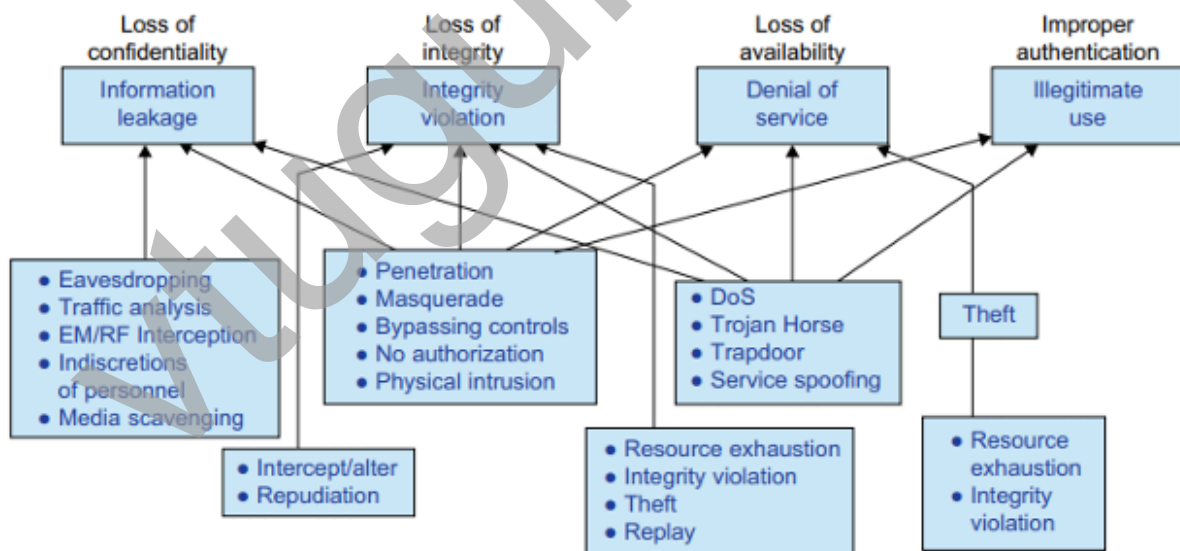


FIGURE 1.25

Various system attacks and network threats to the cyberspace, resulting 4 types of losses.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Loss of Confidentiality

- Occurs when sensitive or private information is revealed without authorization.
- Can result from activities like eavesdropping, analyzing network traffic, or intercepting electromagnetic/radio signals.

Loss of Integrity

- Happens when data is altered, tampered with, or used improperly.
- Caused by attacks such as unauthorized access, impersonation, bypassing security controls, or lack of proper permissions.

Loss of Availability

- Occurs when systems or services are inaccessible to users.
- Often caused by Denial of Service (DoS) attacks, Trojan horse malware, or service spoofing.

Improper Authentication / Unauthorized Use

- Arises when attackers gain access without correct login credentials or permissions.
- Leads to misuse of resources and theft of data due to weak or missing authentication mechanisms.

Module 2

Q.03 a. Demonstrate the architecture of a computer system before and after virtualization.

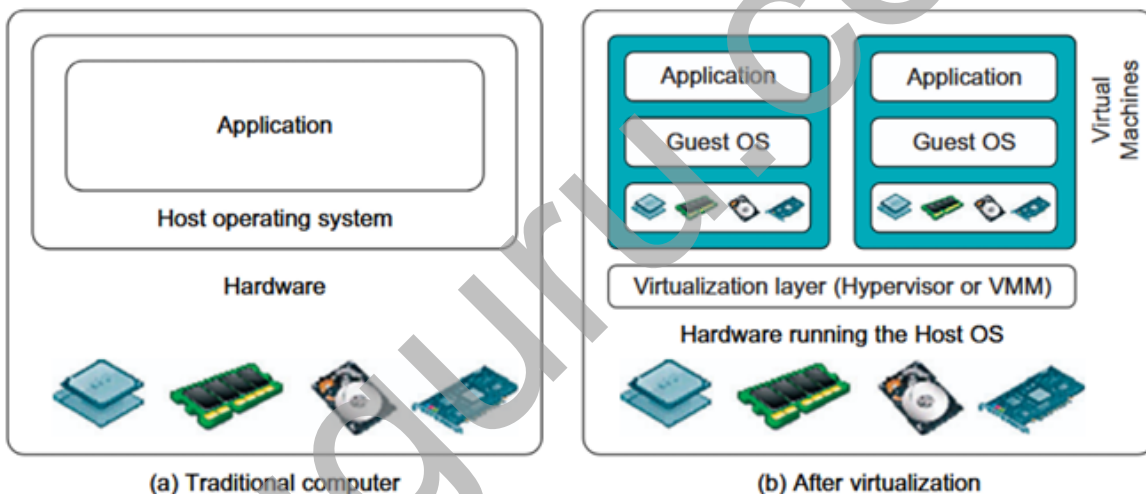
Before Virtualization

- A single operating system is installed directly on the physical hardware.
- All applications depend on this one OS for execution.
- If the OS fails or crashes, every running application also stops.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- Hardware resources are underutilized and efficiency is low.
- It's impossible to run multiple operating systems (like Windows and Linux) simultaneously on the same machine.
- No isolation between applications—one faulty app can disrupt others.
- Difficult to test software in different environments on the same system.
- Installing new applications or operating systems often requires rebooting or reinstalling.
- Not well-suited for cloud computing, data centers, or environments with many users.
- Overall, it offers limited flexibility and is harder to maintain.



After Virtualization

- A hypervisor (Virtual Machine Monitor) operates directly on the hardware.
- Multiple Virtual Machines (VMs) can be created on a single physical server.
- Each VM runs its own operating system and applications independently.
- Failure in one VM does not impact the others.
- Hardware resources are used more efficiently by sharing among VMs.
- Supports running different OSes like Windows and Linux on the same machine.
- VMs can be easily moved, copied, or backed up as needed.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- Well-suited for software testing, cloud computing, and hosting services.
- Helps balance and manage server workloads effectively.
- Offers greater flexibility, cost savings, and simpler management.

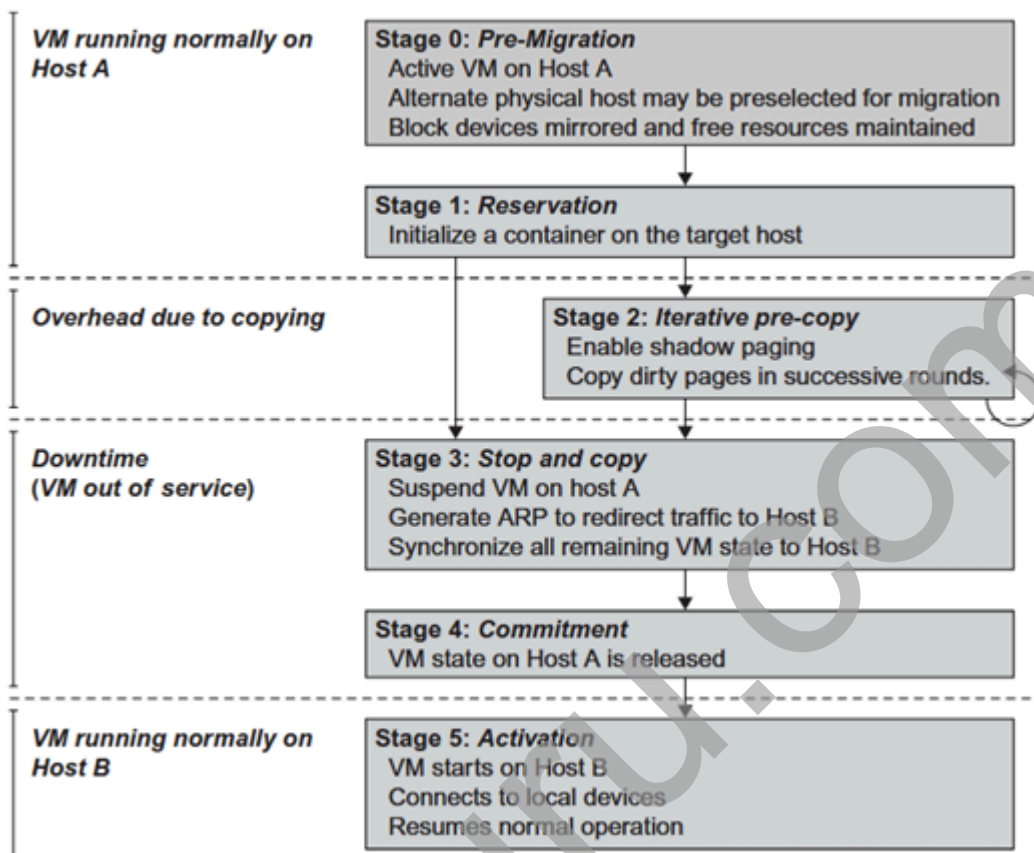
Q.03 b. Compare Physical versus Virtual Clusters.

Sl no	Physical Clusters	Virtual Clusters
1	Built using real, networked physical machines.	Created from virtual machines on physical servers.
2	Each node needs its own dedicated CPU, RAM, and storage.	Multiple VMs share the same hardware resources.
3	More expensive because of additional physical hardware.	Cost-effective since fewer physical servers are needed.
4	Requires significant physical space and higher power.	Saves space and energy by consolidating VMs.
5	Harder to scale—needs manual hardware additions.	Easy to scale by deploying new VMs quickly.
6	Complex and time-consuming to set up and maintain.	Faster setup using virtualization tools.
7	Failure of one node can affect the whole cluster.	Failures are contained; other VMs keep running.
8	Lower efficiency due to idle physical resources.	Higher efficiency through shared hardware use.
9	Moving apps or data between nodes is challenging.	VMs can be migrated easily between servers.
10	Needs manual monitoring and management of each server.	Simplified management with virtualization software like VMware or VirtualBox.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Q.04 a. Construct the Live migration process of a VM from one host to another.



VM States Before Migration

- **Active:** The VM is running normally and handling tasks.
- **Paused:** The VM exists but isn't currently processing anything.
- **Suspended:** The VM's state is saved to disk, making it inactive.

Steps in Live VM Migration

1. Pre-Migration (Step 0):

- VM operates on Source Host A.
- Target Host B is chosen and prepared for migration.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

2. Reservation (Step 1):

- A VM container is created and set up on Host B.

3. Iterative Pre-Copy (Step 2):

- VM memory is transferred to Host B in multiple rounds.
- Modified memory pages (dirty pages) are re-copied.
- The VM keeps running during this process.

4. Stop and Copy (Step 3):

- VM on Host A is paused.
- Final memory state, CPU, and network data are sent to Host B.
- Network connections are redirected to Host B.
- This brief pause is called downtime.

5. Commit (Step 4):

- Resources are released on Host A.
- Host A no longer has control over the VM.

6. Activation (Step 5):

- VM is launched on Host B.
- It resumes operation and connects to local resources.

Q.04 b. Develop a architecture of livewire for Intrusion Detection using a dedicated VM.

Intrusion and Intrusion Detection Systems (IDS)

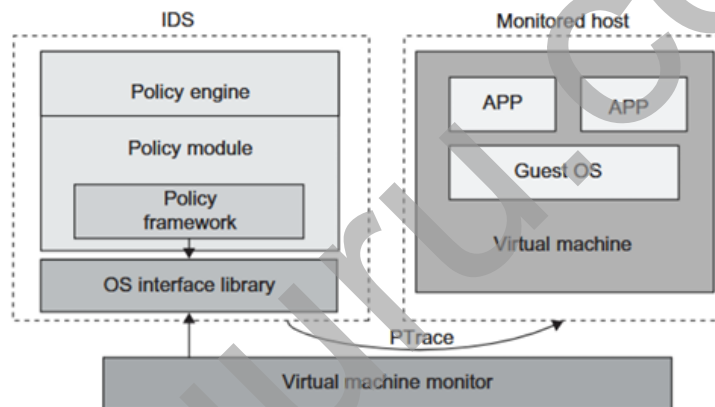
- *Intrusion* refers to unauthorized access to a computer system by local users or over a network.
- An *Intrusion Detection System (IDS)* is designed to identify and alert on such unauthorized activities.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Types of IDS:

- **Host-based IDS (HIDS):**
 - Runs directly on the machine it monitors.
 - Vulnerable if the host system itself is compromised.
- **Network-based IDS (NIDS):**
 - Observes network traffic to detect suspicious activity.
 - May struggle to recognize spoofed or disguised attacks.



IDS in Virtualized Environments:

- In virtual systems, guest VMs are isolated from one another.
- If one VM is compromised, others remain safe—similar to how NIDS works by protecting overall traffic.
- The Virtual Machine Monitor (VMM) watches access requests, functioning like a HIDS to trace suspicious activities.

VM-based IDS Approaches:

1. Runs as a separate monitoring process in each VM or in a dedicated high-privileged VM.
2. Built directly into the VMM for complete hardware-level access and monitoring.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Key Components of a VM-based IDS:

- **Policy Engine:** Evaluates and enforces security rules.
- **Policy Module:** Utilizes tools like PTrace to monitor and apply security controls inside guest VMs.

Post-Attack Considerations:

- Completely preventing intrusions is challenging, so analyzing attacks after they occur is critical.
- Logs help trace attack behavior, but they can't always be trusted if the operating system is compromised.

Honeypots and Honeynets:

- Special systems are set up to lure attackers away from real targets.
- Can be either physical machines or virtual instances.
- Virtual honeypots must be carefully controlled to ensure they can't be used to attack the host or the VMM.

Module 3

Q.05 a. Outline six design objectives for cloud computing.

Key Cloud Computing Trends and Requirements

1. Transition from Desktop to Data Center:

- Computing power, storage, and software move from individual desktops to centralized data centers accessible over the Internet.

2. Service Delivery and Cloud Economics:

- Cloud services are offered under Service Level Agreements (SLAs) with users.
- Follows a pay-as-you-go pricing model while aiming for efficient use of power and resources.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

3. Scalable Performance:

- Cloud systems need to handle increasing numbers of users by scaling performance as demand grows.

4. Protecting Data Privacy:

- Users must trust that cloud providers will keep their private data secure and confidential.

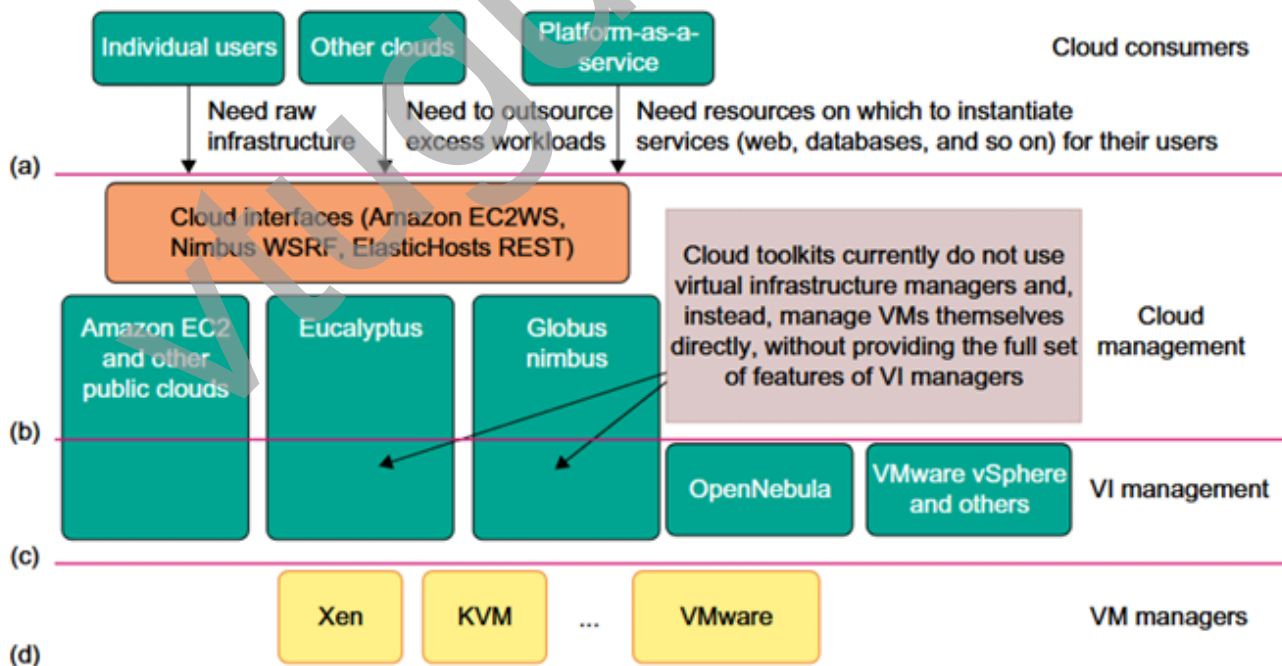
5. Maintaining High-Quality Services:

- Quality of Service (QoS) needs to be standardized so services work smoothly across different providers.

6. Developing New Standards and Interfaces:

- Universal APIs and access protocols are required to avoid vendor lock-in and allow easy movement of applications between different cloud platforms.

Q.05 b. With a neat diagram, build a cloud ecosystem with a private cloud.



BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Cloud Ecosystem Overview

- A cloud ecosystem brings together cloud service providers, end-users, and technologies to work in a connected environment.
- Public clouds are widely adopted and serve as the foundation of most cloud infrastructures.
- Organizations often use private and hybrid clouds to combine internal systems with public cloud services.
- Users expect flexible platforms that support applications like websites and databases.
- Cloud management systems deliver virtualized resources through Infrastructure as a Service (IaaS) platforms.
- Virtual infrastructure managers assign virtual machines across multiple server clusters for efficient use.
- VM management software such as Xen, KVM, and VMware is used to operate virtual machines on physical hardware.
- Tools like OpenNebula, vSphere, Nimbus, and Eucalyptus help monitor and control cloud environments.
- Many startups rely on cloud services rather than building their own in-house IT infrastructure.
- Interfaces like Amazon EC2 Web Services (EC2WS), Nimbus WSRF, and ElasticHosts REST provide access to cloud features.
- Virtual infrastructure tools also support load balancing, auto-scaling, and optimal resource utilization.

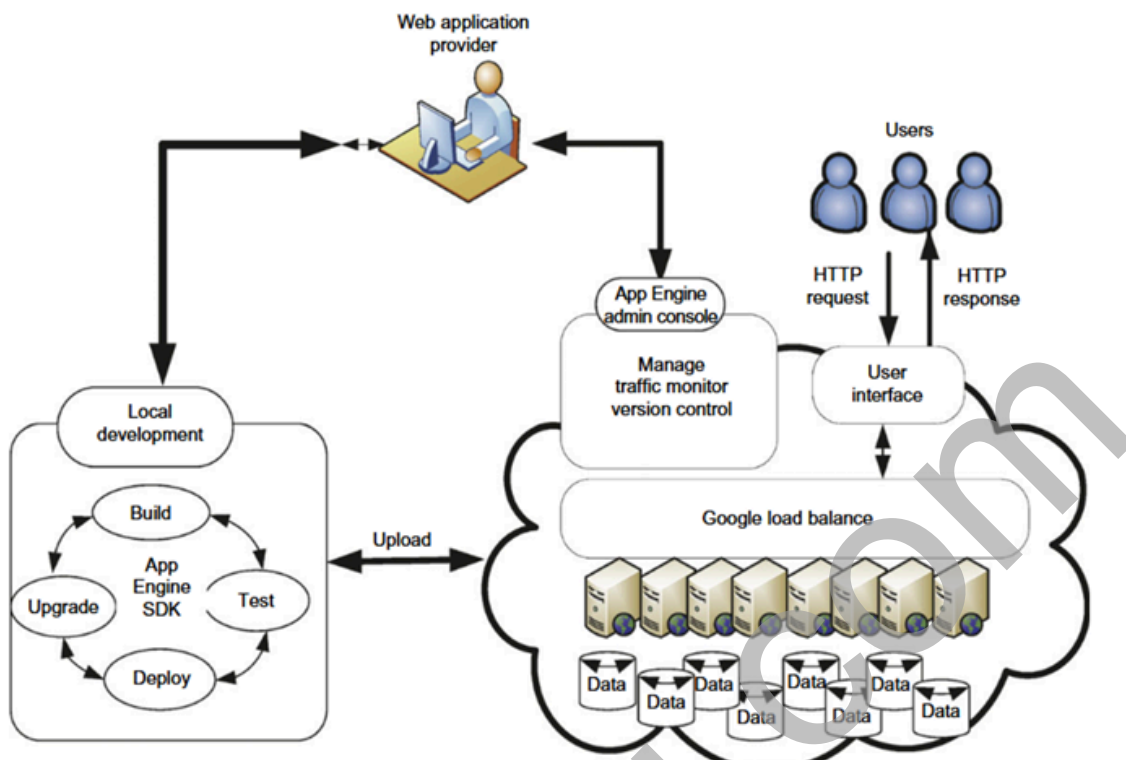
Q.05 c. Organize Functional Modules of GAE.

Google App Engine (GAE) Overview

- Google App Engine is a Platform-as-a-Service (PaaS) that enables developers to create and deploy web applications on Google's cloud infrastructure.
- It offers several key components to simplify application development and hosting.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University



Main Functional Modules of GAE:

1. Runtime Environment:

- Supports applications written in languages like Java, Python, Go, and PHP.

2. Datastore:

- A NoSQL database service used to store structured application data.

3. Task Queues:

- Manages background processing tasks without interrupting the main user flow.

4. Memcache:

- Offers a fast, in-memory cache to store frequently accessed data and reduce latency.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

5. User Authentication:

- Provides built-in APIs to handle user logins and identity management.

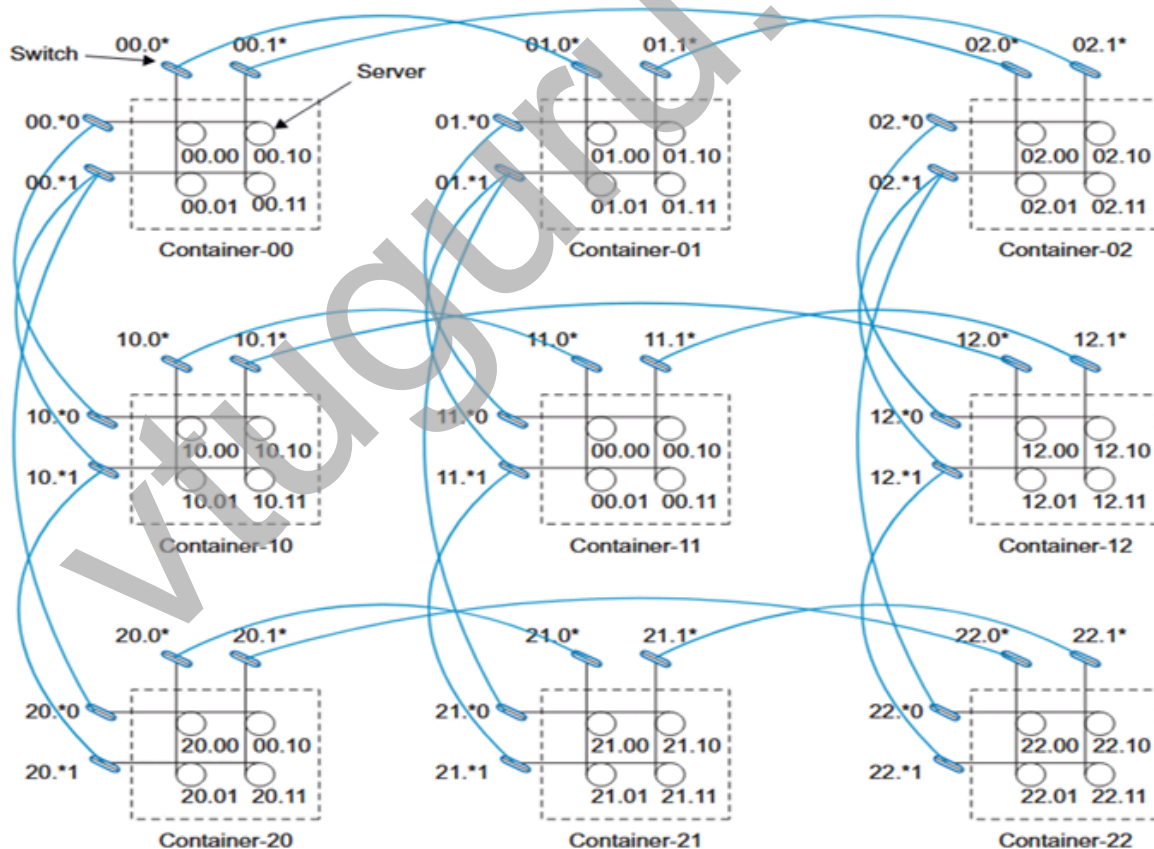
6. App Versioning and Deployment:

- Allows developers to deploy multiple versions of their application and easily switch or roll back between them.

Q.06 a. Identify basic requirements for managing the resources of a data center.

Data Center Management: Key Challenges and Considerations

Managing a data center involves ensuring that all operations run securely, efficiently, and without interruption. These challenges are based on real-world experiences in the IT and cloud computing industries.



BIS613D Solved Model Question Paper

Visvesvaraya Technological University

User Satisfaction:

The infrastructure must consistently deliver high-quality service over a long period—ideally up to 30 years. Maintaining strong Quality of Service (QoS) is essential.

Controlled Information Flow:

Data must move seamlessly between systems to ensure smooth operations. High availability (HA) and uninterrupted service are critical for user trust and system functionality.

Multiuser Support:

A modern data center should be capable of handling multiple users simultaneously. This includes tasks like managing traffic, updating databases, and monitoring servers.

Scalability:

As demand grows, the system must be able to expand its resources. This includes storage, processing capabilities, power supply, input/output (I/O) systems, and cooling infrastructure.

Reliability in Virtual Environments:

Features like fault tolerance, automatic failover, and live migration of virtual machines are vital for quick recovery from hardware failures or natural disasters.

Cost Efficiency:

The overall cost of managing a data center—including hardware, electricity, staffing, and maintenance—must be optimized to benefit both the cloud provider and the end users.

Security and Data Protection:

Robust security systems are needed to guard against cyberattacks and insider threats. Protecting data privacy and maintaining integrity is a top priority.

Green Computing and Energy Efficiency:

Energy-saving systems and eco-friendly infrastructure help reduce power consumption and environmental impact, while also lowering operational expenses.

Service Automation:

Automated tools should be used for repetitive tasks like load balancing, backups, and software updates. This improves efficiency and minimizes human error.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Monitoring and Reporting:

Continuous monitoring of system performance and usage is essential. Detailed reports help identify problems, plan upgrades, and ensure smooth day-to-day operations.

Q.06 b. Summarize six open challenges in cloud architecture development.

Challenge 1: Service Availability and Data Lock-in

- When a cloud provider experiences downtime, the entire system can fail, especially if it's managed by a single company.
- Using multiple providers can help improve overall service availability and reduce risk.
- Proprietary APIs often lead to "lock-in," making it difficult for users to move their applications or data between different cloud platforms.

Challenge 2: Data Privacy and Security Concerns

- Cloud systems are exposed to cyber threats such as DDoS attacks, malware infections, and virtual machine hijacking.
- Data theft or misuse can occur if it's not properly encrypted and secured.
- Legal requirements in some countries demand that certain data stays within national borders, adding complexity.

Challenge 3: Unpredictable Performance and Bottlenecks

- While VMs share CPU and memory efficiently, I/O operations like disk access can create slowdowns.
- Large-scale applications often suffer from network traffic delays and transfer issues.
- Preventing bottlenecks requires improved VM placement strategies and hardware upgrades.

Challenge 4: Distributed Storage and Debugging Issues

- Cloud systems need flexible storage solutions that can scale up or down based on demand.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- Debugging problems in large-scale systems is difficult because many bugs appear only when the system is fully loaded.
- Virtual machines and simulation tools are helpful for collecting detailed debugging information.

Challenge 5: Scalability, Interoperability, and Standards

- Cloud services must scale quickly to handle more users while still meeting SLAs.
- Standard VM formats like OVF help applications run on various platforms.
- However, cross-platform migrations between different CPU architectures (e.g., Intel and AMD) remain challenging.

Challenge 6: Software Licensing and Reputation Management

- Cloud environments need adaptable licensing models, such as pay-per-use or bulk options.
- Malicious activity from one user can harm the entire cloud provider's reputation, for example by causing IP blacklisting.
- Legal responsibilities between providers and customers should be clearly defined in Service Level Agreements (SLAs).

Q.06 c. Summarize six open challenges in cloud architecture development.

Amazon Web Services (AWS) Overview

1. Infrastructure as a Service (IaaS):

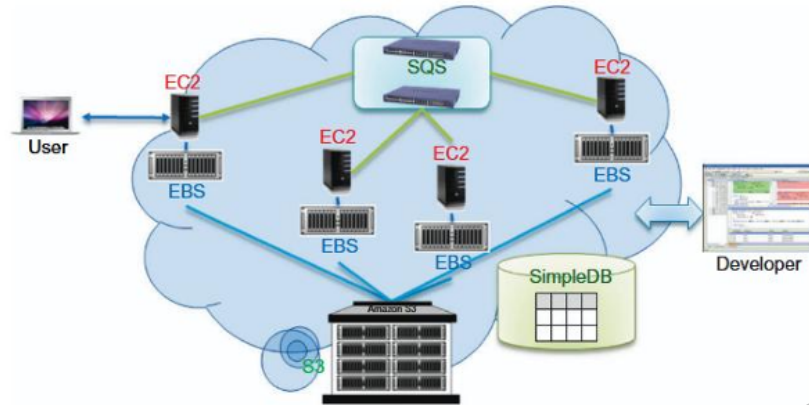
- AWS uses the IaaS model through EC2 (Elastic Compute Cloud), providing virtual machines for running cloud applications.

2. Storage Solutions (S3 and EBS):

- **S3 (Simple Storage Service):** Offers scalable object-based storage for files, backups, and media.
- **EBS (Elastic Block Store):** Provides block-level storage suitable for applications that need consistent, reliable performance.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University



3. Messaging Services (SQS and SNS):

- **SQS (Simple Queue Service):** Stores and manages messages even if the receiver is offline, ensuring reliable communication.
- **SNS (Simple Notification Service):** Delivers notifications and alerts to applications, users, or other systems.

4. Performance and Monitoring Tools:

- **ELB (Elastic Load Balancer):** Evenly distributes incoming traffic across multiple EC2 instances for improved reliability.
- **CloudWatch:** Monitors CPU, memory, and network use to help manage resources effectively.
- **Auto Scaling:** Automatically adjusts the number of EC2 instances based on demand to maintain performance and control costs.

5. Database Services:

- **RDS (Relational Database Service):** Provides managed relational databases like MySQL, PostgreSQL, and Oracle with automated backups and scaling.

6. Content Delivery:

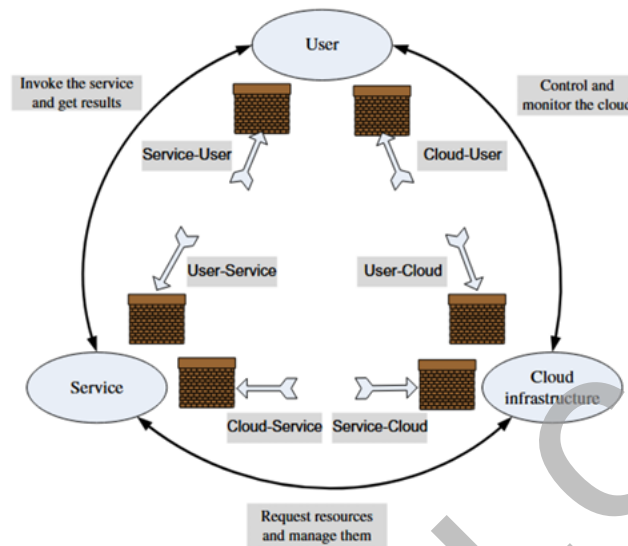
- **CloudFront:** A content delivery network (CDN) that securely delivers data, videos, and applications to users globally with low latency.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Module 4

Q.07 a. Demonstrate surfaces of attacks in a cloud computing environment with neat diagram.



Cloud Security Risks and Threats

1. Easy Onboarding Increases Risk:

- Many users adopt cloud services quickly without fully understanding the security implications or ethical considerations.
- This makes systems more vulnerable to misuse or unsafe practices.

2. Clouds as Tools for Large-Scale Attacks:

- Cloud infrastructure can be exploited to launch massive cyberattacks against other systems.
- Preventing such abuse is a critical challenge for providers.

3. Three Main Categories of Cloud Risks:

- (i) Traditional security threats.
- (ii) Availability and reliability issues.
- (iii) Risks related to third-party control over user data.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- Without careful management, these risks make cloud computing less secure.

4. Traditional Security Threats:

- Examples include DDoS attacks, phishing, SQL injection, and cross-site scripting.
- Because cloud resources are shared, a single attack can impact many users at once.

5. Challenges in Authentication and Access Control:

- Different users within the same organization may need varying levels of access.
- Aligning a company's internal security policies with those of the cloud provider can be complex.

6. Difficulty in Tracing Attacks:

- Identifying how an attack occurred in a cloud environment is challenging.
- Traditional investigation tools, like system logs, may not work effectively in cloud systems.

7. Service Availability Concerns:

- Events like power outages, hardware failures, or natural disasters can interrupt cloud services.
- Companies relying heavily on the cloud may face serious problems if their data becomes inaccessible.

8. Trust Issues with Third-Party Vendors:

- Cloud providers may rely on external vendors or substandard hardware.
- Users lose visibility and control, which can lead to data loss or compromised security.

9. Limited Provider Responsibility:

- Many providers, like AWS, disclaim liability for data loss or service failures in their terms of use.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- This means users often bear the risk without strong guarantees.

10. Top Seven Threats (CSA Report 2010):

- Abuse of cloud services.
- Insecure APIs.
- Malicious insiders.
- Shared technology vulnerabilities.
- Account hijacking.
- Data loss or leakage.
- Unknown risk profile.
- While IaaS is exposed to all these threats, PaaS and SaaS models may face fewer of them.

Q.07 b. List out the top cloud security threats of CSA2016.

Cloud Security Threats Explained

1. Data Breaches:

- Occur when unauthorized individuals gain access to sensitive or confidential information.

2. Weak Identity, Credential, and Access Management:

- Poor password practices or stolen login details can lead to unauthorized system access.

3. Insecure APIs (Application Programming Interfaces):

- Vulnerable or poorly designed APIs can be exploited by attackers to bypass security controls.

4. System Vulnerabilities:

- Software bugs or flaws can be used by attackers to compromise systems and data.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

5. Account Hijacking:

- Attackers gain control of user accounts and services using stolen credentials.

6. Malicious Insiders:

- Employees, contractors, or partners with authorized access intentionally misuse their privileges.

7. Advanced Persistent Threats (APTs):

- Sophisticated, long-term targeted attacks aimed at stealing sensitive data or disrupting operations.

8. Data Loss:

- Permanent loss of data due to accidental deletion, hardware failure, or inadequate backup strategies.

9. Insufficient Due Diligence:

- Organizations moving to the cloud without fully understanding security responsibilities or risks involved.

10. Abuse and Misuse of Cloud Services:

- Attackers use cloud infrastructure to send spam, host malware, or launch DDoS attacks.

11. Denial of Service (DoS) Attacks:

- Attackers overwhelm cloud services, preventing access for legitimate users.

12. Shared Technology Vulnerabilities:

- Risks arising from the multi-tenant model where multiple users share the same underlying infrastructure.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Q.07 c. Select four widely-accepted fair information practices that “Consumer oriented commercial web sites that collect personal identifying information from or about consumers online would be required to comply with.

Fair Information Practices

1. Notice:

- Websites should clearly inform users about their data collection policies.
- This includes details about what data is collected, the methods used (such as cookies), how the data is used, and whether it is shared with other organizations.

2. Choice:

- Users must be given the option to decide how their personal information is used.
- This applies to both internal purposes like marketing and external sharing with third-party partners.

3. Access:

- Users should have the ability to view, correct, or delete their own personal information.
- This ensures greater transparency and gives users control over their data.

4. Security:

- Websites need to take appropriate measures to safeguard user data against theft or misuse.
- Security practices should be flexible and adaptable to support evolving technology and threats.

Q.08 a. Summarize The design goals of Xoar are.

- ❖ Xoar is an enhanced version of the Xen hypervisor that uses microkernel design principles to strengthen security.
- ❖ It operates under the assumption that system administrators are trusted, while the main risks come from guest virtual machines or vulnerabilities in management code.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- ❖ Xoar retains all standard Xen features but implements strict privilege separation, giving each component only the permissions it truly needs.
- ❖ Interfaces are kept minimal to limit the attack surface, and any sharing between components is either avoided or carefully logged.
- ❖ Components are activated only when necessary, reducing the window of opportunity for attacks.
- ❖ Xoar also supports secure audit logging, making it easier to trace and investigate security events.

Types of System Components in a Secure Cloud Architecture

Cloud systems often classify components based on when and how they operate. The four main types are:

1. Always Running Components:

- These are continuously active and essential to the system's operation.
- *Example:* XenStore-State, which helps manage VM state information.

2. Boot-Time Only Components:

- These are required during system startup and are removed or shut down once booting is complete.

3. On-Demand Components:

- These are loaded only when needed for a specific task or service, helping reduce memory usage and potential attack surfaces.

4. Timer-Based Components:

- These are restarted or activated at regular intervals using a system timer, often used for routine checks or updates.

- ❖ A modular approach lowers the system's overall risk and security footprint while causing only minor performance overhead.
- ❖ This design breaks functionality into separate components, each handling a specific task.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Examples of such modules include:

- **Builder:** Responsible for launching new virtual machines.
- **QEMU:** Provides device emulation to support various hardware interfaces.
- **PCIBack and NetBack:** Specialized drivers that manage PCI device access and network communication for guest VMs.

Q.08 b. Explain mobile devices and cloud security.

1. Mobile Cloud Ecosystem:

- Mobile applications often rely on cloud services for storing data, creating backups, and handling processing tasks.
- This is necessary because smartphones and tablets have limited CPU power, memory, and storage capacity.

2. Security Challenges on Mobile Devices:

- Mobile devices commonly connect through public or unsecured Wi-Fi networks, making them vulnerable to interception by attackers.
- They are also easily lost or stolen, increasing the chances of unauthorized access to sensitive data.

3. Authentication and Identity Management:

- Weak or reused passwords on devices can allow attackers to gain access to cloud accounts.
- Best practices include using multi-factor authentication (MFA) to strengthen security.

4. Data Protection During Transit and Storage:

- Data should be encrypted while being transmitted (using protocols like TLS/SSL) and when stored in the cloud.
- End-to-end encryption ensures that only authorized users on the device can read the data.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

5. Application and API Security:

- Malicious or insecure apps can access or leak user data stored in the cloud.
- Secure mobile applications require reliable APIs with strong access control measures.

6. Device and Application Management:

- Organizations often use Mobile Device Management (MDM) tools to enforce encryption, require strong pass codes, deliver updates, and enable remote data wiping.
- These measures help protect corporate data even if a device is lost or stolen.

Q.08 c. Model an overview of reputation system design options.

Types of Reputation System Models

1. Centralized Reputation System:

- A single authority is responsible for collecting, evaluating, and maintaining reputation scores.
- While easy to monitor and manage, it can become a single point of failure or a target for attacks.

2. Distributed Reputation System:

- Reputation information is shared across multiple peers or systems.
- This avoids a central control point but makes it harder to ensure consistency and trust across the network.

3. User-Based Ratings:

- Users directly provide feedback, such as stars, likes, or reviews, after using a service.
- Simple to implement but vulnerable to manipulation through fake reviews or Sybil attacks.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

4. Behavior-Based Monitoring:

- The system assesses actual performance metrics like uptime, response times, and data accuracy.
- Offers more objective and reliable scoring but requires sophisticated tracking and analysis tools.

5. Context-Aware Reputation:

- Reputation is tailored to specific services or environments, such as reliability in storage versus computation tasks.
- Supports more precise trust decisions suited to different scenarios.

6. Time-Based Reputation:

- Reputation scores decline over time if not maintained or updated.
- This approach promotes ongoing good behavior and discourages users from building trust and then acting maliciously.

7. Incentive-Driven Models:

- Users receive rewards like credits or higher trust scores for honest feedback or good behavior.
- Encourages truthful reporting but must be carefully designed to prevent misuse of the incentives.

Module 5

Q.09 a. Outline Important Cloud Platform Capabilities.

On-Demand Self-Service:

- Users can access computing resources such as servers and storage whenever they need them, without needing help from service providers.

Broad Network Access:

- Cloud services are available over the Internet and can be used on various devices like laptops, smartphones, and tablets.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Resource Pooling:

- Providers share resources like storage and memory across many users through virtualization, improving efficiency.

Rapid Elasticity:

- Resources can be scaled up or down quickly based on demand using features like auto-scaling.

Measured Service (Pay-as-You-Go):

- Users are charged only for the resources they actually use, making it cost-effective—similar to prepaid mobile plans.

High Availability:

- Cloud platforms ensure services are available around the clock (24/7) by using backups and load balancing.

Security and Compliance:

- Includes features like user authentication, data encryption, firewalls, and adherence to government regulations to keep data safe.

Automation:

- Tasks such as backups, software updates, and scaling can be handled automatically, reducing manual effort.

Multi-Tenancy:

- Allows multiple users to securely and privately share the same cloud infrastructure.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

APIs and Developer Tools:

- Provides easy-to-use tools for developers to build, test, and deploy applications on the cloud platform.

Q.09 b. Organize the steps involved in MapReduce.

MapReduce Processing Steps

1. Input Splitting:

- The large input dataset is divided into smaller chunks or blocks so it can be processed in parallel.

2. Map Function:

- Each block is processed independently using the Map function.
- The data is transformed into key-value pairs.
- *Example:* The word "apple" becomes `<apple, 1>`.

3. Shuffling:

- The system reorganizes data so that all values with the same key are grouped together.
- *Example:* All `<apple, 1>` pairs are brought to the same reducer.

4. Sorting:

- The keys are sorted before they are passed to the Reduce function to ensure proper grouping and ordering.

5. Reduce Function:

- The grouped key-value pairs are processed to generate meaningful results.
- Operations such as counting, summing, or averaging are performed.
- *Example:* `<apple, [1,1,1]>` is reduced to `<apple, 3>`.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

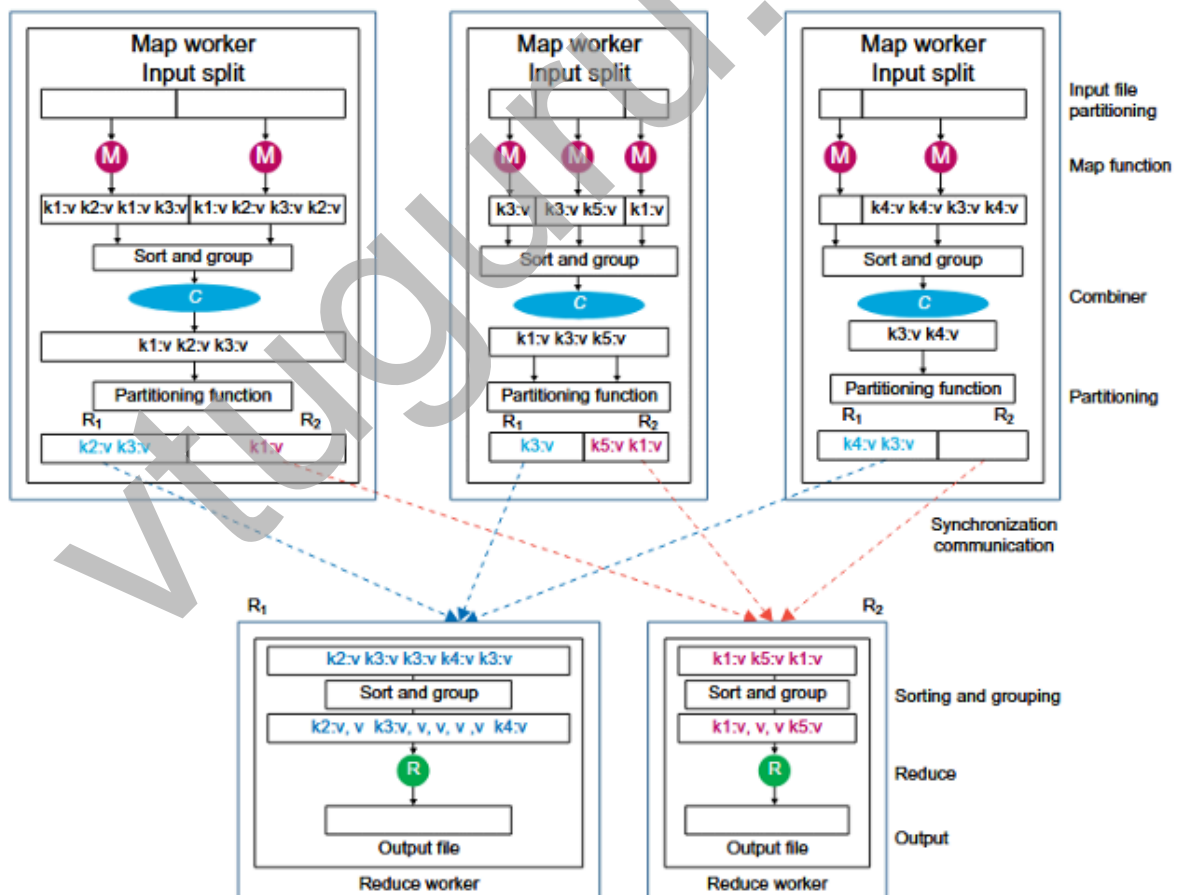
6. Output Generation:

- The final results are stored in the file system in key-value format for further use.

Simple Example:

- **Input:** List of words → ["apple", "apple", "banana"]
- **Map Output:** <apple, 1>, <apple, 1>, <banana, 1>
- **Shuffle and Reduce Output:** <apple, 2>, <banana, 1>

Q.10 a. Explain with a neat diagram how data flows in running a MapReduce job at various task trackers using the Hadoop library.



BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Phase 1: Setup and Input Preparation

1. Data Partitioning:

- The input file stored in HDFS is divided into multiple chunks called Input Splits.
- Each split is assigned to a separate Map Task for processing.

2. Computation Partitioning:

- The developer provides custom Map() and Reduce() functions.
- The Hadoop framework runs these user-defined programs on worker nodes.

3. Master and Worker Setup:

- One node acts as the Master (JobTracker), while others serve as Workers (TaskTrackers or NodeManagers).
- The Master is responsible for assigning Map and Reduce tasks to the Worker nodes.

Phase 2: Map-Side Processing

4. Input Reading:

- Each Map worker reads its assigned input split and feeds it into the Map() function.

5. Map Function Execution:

- Generates intermediate output in the form of key-value pairs.

6. Combiner (Optional):

- Performs local aggregation (like summing values) to reduce the amount of data sent over the network.

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

7. Partitioning Function:

- Prepares data to be divided among different Reduce tasks, typically using a hash function.

Phase 3: Shuffle and Reduce

8. Synchronization:

- Intermediate data is divided into R partitions (one for each Reduce task) using a formula like $\text{Hash}(\text{key}) \bmod R$.
- Reduce workers wait for all Map tasks to finish before starting.

9. Communication:

- Reduce workers fetch their assigned partitions from all Map workers via Remote Procedure Calls (RPC).

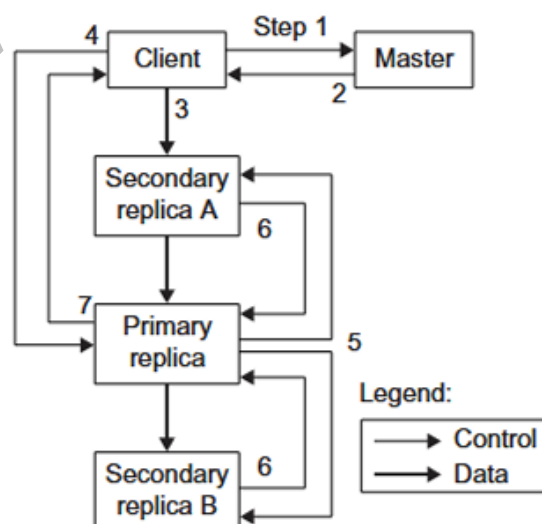
10. Sorting and Grouping:

- The Reduce side sorts keys and groups all values that share the same key together.

11. Reduce Function Execution:

- Processes the grouped data and writes the final output results to files in HDFS.

Q.10 b. Explain Data mutation sequence in GFS with diagram1.



BIS613D Solved Model Question Paper

Visvesvaraya Technological University

Steps in GFS (Google File System) Write Operation

1. Client Requests Chunk Information:

- The client contacts the Master server to find out which chunk server holds the lease for the desired chunk and the locations of other replicas.

2. Master Server Responds:

- The Master replies with the identity of the primary replica and the addresses of the secondary replicas.
- The client stores (caches) this information for later use.

3. Client Sends Data to Replicas:

- The client pushes the data to all replicas (both primary and secondaries).
- Each chunk server stores the data temporarily in its buffer cache.
- This process is separated from control flow to improve performance.

4. Client Initiates Write with Primary:

- Once all chunk servers have received the data, the client sends a write request to the primary replica to start the actual write operation (mutation).
- The primary assigns serial numbers to maintain the correct write order.

5. Primary Coordinates with Secondaries:

- The primary sends write commands to all secondary replicas in the same order (based on the serial numbers) to ensure consistency.

6. Secondaries Send Acknowledgment:

- Each secondary replica confirms that the data has been written successfully.

7. Primary Confirms to Client:

- After receiving confirmations from all secondaries, the primary sends a final success response to the client.
- If any failure happens during this process, the client:

BIS613D Solved Model Question Paper

Visvesvaraya Technological University

- Marks the write as failed.
- Either retries the write (steps 3–7) or restarts the operation from the beginning (step 1), depending on the error.

Vtuguru.com